Nhill 3.0 — Installation guide

Français : ../fr/guide-installation.md

This guide covers installing the **applications** (Linux, Windows, Android), the **server** (Linux) and upgrading from 2.x. Building the packages: build-guide.md. Settings: configuration-guide.md. Marketing website: `website/README.en.md`.

> **State of the deliverables.** Every file shipped in `release/V<version>/` was actually built; a missing platform is listed in `release/V<version>/checksums/MISSING.txt`. The validation report (validation-report.md) says what was tested and on which machine. The Windows installers and the APK are **not signed** with a publisher certificate until the creators acquire one.

1. Verify a downloaded file

Before installing, compare the file's SHA-256 digest with the one in `checksums/SHA256SUMS` (published separately from the file, for example on the website):

```
sha256sum -c SHA256SUMS --ignore-missing          # Linux
```

```
Get-FileHash .\nhill-business-3.0.0-setup.exe -Algorithm SHA256   # Windows: compare by eye
```

If the digest differs, do not install the file.

2. Minimum configuration

Component	System	Memory	Notes
Nhill Business, POS, Console (desktop)	Windows 10/11 64-bit; Debian 13 or Ubuntu 24.04 (amd64)	to be measured on the target computer	1280×720 screen or larger recommended; POS works from 860×560
Nhill Business (Android)	Android 7.0 (API 24) or later, arm64 processor	—	Internet connection required
Server	Debian 13 or Ubuntu 24.04, amd64	starting point: 2 vCPU, 4 GB, 40 GB SSD	PostgreSQL 15+ (Supabase or self-hosted), synchronised clock (NTP)

The client computers' memory needs have not been measured: test on the hardware actually used (see the validation report).

3. Applications

3.1 Linux (Debian 13, Ubuntu 24.04)

Packages are built **for one specific distribution**: take those from your system's folder (`linux/debian-13/` or `linux/ubuntu-24.04/`).

```
sudo apt install ./nhill-business_3.0.0-1_amd64.deb ./nhill-pos_3.0.0-1_amd64.deb
sudo apt install ./nhill-console_3.0.0-1_amd64.deb     # for the creators of Nhill only
```

`apt` installs the required libraries from the official repositories. The applications appear in the menu (Office). Remembered sessions need a Secret Service keychain (GNOME Keyring or KWallet), present on usual desktops.

Removal: `sudo apt remove nhill-business nhill-pos nhill-console`.

3.2 Windows 10/11

1. Check the digest (§ 1), then run `nhill-business-3.0.0-setup.exe` (or `nhill-pos-…`, `nhill-console-…`). Installation asks for administrator rights (it installs into `Program Files`).
2. As long as the installer is unsigned, Windows SmartScreen may show “Windows protected your PC”. Bypass this warning **for this file only** and **only if the digest matches**: “More info” → “Run anyway”. Never turn off SmartScreen or the antivirus.
3. Start-menu shortcuts are created; remove through Settings → Apps.

Silent installation (fleet of computers): `nhill-pos-3.0.0-setup.exe /S`.

3.3 Android

1. Check the digest of `nhill-business-3.0.0-arm64-v8a.apk`.
2. Open the file on the phone; Android asks you to allow **this app** (file manager or browser) to install apps. Allow it for this installation, then withdraw the permission (Settings → Apps → Special access → Install unknown apps).
3. An APK signed with the **debug** key (built without a release key) can be installed for trials but must not be distributed to customers; it cannot be upgraded by an APK signed with the release key (uninstall first).

Nhill POS and Nhill Console are not offered on Android (desktop applications).

3.4 First start

No server address is compiled into the application. Nhill Business asks for it when a business is created; Nhill Console and Nhill POS ask for it at their first sign-in. Enter the address given by your provider (HTTPS): it is checked, then saved on the device, and can be changed in the settings. For a one-off trial on another server, the `NHILL_SERVER` environment variable replaces it (desktop only). Then: nhill-business.md § 1, nhill-pos.md § 2, nhill-console.md § 2.

4. Server (production)

> Procedure chosen for production: **Hostinger VPS in France**, script `deploy/vps/provision.sh` — see vps-deployment-hostinger.md. This section describes the manual installation, useful for another hosting.

Target: a Linux server for the API, a PostgreSQL 15+ database (**Supabase** or self-hosted), a domain name and a TLS proxy (Caddy). The full procedure has not yet been run on a public production server with TLS: run it on a staging server first.

4.1 Prerequisites

- Synchronised clock (NTP) — essential for TOTP and webhooks.
- Firewall: only 22 (administration, restricted) and 80/443 (Caddy) open.
- DNS: `api.<your-domain>` → server address.

4.2 Package

```
sudo apt install ./nhill-server_3.0.0-1_amd64.deb
curl -s http://127.0.0.1:8080/health     # {"database":false,"status":"setup"}
```

The package creates the system users `nhill` and `nhill-backup`, the file `/etc/nhill/server.env` (mode 600) with an `NH_SECRET_KEY` **generated on the server** and no database, then starts the service in **setup mode**. Copy `NH_SECRET_KEY` to a vault immediately (without it, establishment codes and the administrators' TOTP secrets become unreadable).

4.3 Database

- **Supabase**: Connect → **Session pooler** (port 5432, user `postgres.<reference>`); not the transaction pooler (6543). No API key (`anon`, `service_role`) is needed or accepted; Nhill creates a `nhill` schema closed to Supabase's API roles.
- **Self-hosted PostgreSQL**: empty `nhill` database, an account able to create roles, `scram-sha-256` in `pg_hba.conf`, TLS if the database is remote.

4.4 Installation from Nhill Console

1. Setup token (60 minutes, single use): `journalctl -u nhill-server | grep -i -A2 "setup mode"`.
2. Nhill Console → **Install a new server** → address and token → database type, host, port, user, password → **Test the connection** → first administrator → **Initialise the database**.
3. The server creates `nhill_owner`, `nhill_app`, `nhill_backup` (random passwords), applies the migrations, checks isolation, writes `/etc/nhill/server.env` and restarts. The database administration password is not kept; the backup account's password is shown **only once**.

Check: `/health` → `{"database":true,"status":"ok"}`. Command line: `NH_SETUP_DB_PASSWORD=… sudo -u nhill nhill-server --config /etc/nhill/server.env setup --host … --user … --admin-email … --admin-name "…"`.

4.5 TLS and settings

```
sudo apt install caddy
sudo cp /usr/share/nhill/examples/Caddyfile.example /etc/caddy/Caddyfile   # replace
api.example.com
sudo systemctl reload caddy
curl -sI https://api.<domain>/health       # 200 + Strict-Transport-Security
```

In `/etc/nhill/server.env`: `NH_PUBLIC_BASE_URL=https://api.<domain>`, `NH_TRUST_PROXY=1`; then `sudo systemctl restart nhill-server` and `sudo -u nhill nhill-server --config /etc/nhill/server.env check`.

4.6 Backups, first access, plans

- Encrypted backups (age): database.md § 5; `sudo systemctl enable --now nhill-backup.timer`, then **a restore test**.
- Nhill Console → **First access** (30-minute token, password of at least 8 characters, mandatory TOTP).
- **Plans and prices**: no price is preconfigured.
- **Mail**: SMTP relay with TLS (`NH_EMAIL_TRANSPORT=smtp`, `NH_SMTP_*`); without it, owners and administrators cannot recover a forgotten password.
- **Payment providers**: stay in test mode until a successful test payment with the provider's account.

5. Upgrading to 3.0.0

From 2.1.x (or 2.0.x):

1. **Back up** the database (on the VPS, `nhill-update` does it itself; elsewhere: `sudo systemctl start nhill-backup.service`) and keep a copy of `NH_SECRET_KEY`.
2. Server: `sudo apt install ./nhill-server_3.0.0-1_amd64.deb` (on the VPS: `sudo nhill-update nhill-server_3.0.0-1_amd64.deb --sha256sums SHA256SUMS`). The server applies migrations **0012 to 0015** (configurable plans and rights, document notices and till invoices, erasure of a business, searches). Existing plans keep their limits (users, tills, warehouses); check each plan's features and limits in Nhill Console › Plans & prices.
3. `sudo -u nhill nhill-server --config /etc/nhill/server.env check`, then `/health`.
4. Applications: install the 3.0.0 versions. They ask for the server address at first launch (no address is built in any more): give it to the users.
5. Optional: `NH_SUPPORT_EMAIL` (support address in the signature of automatic e-mails).

Rollback: a 2.x server refuses a database migrated by 3.0; the only method is to reinstall the previous package **and** restore the database copy taken in step 1 (`nhill-rollback` does it on the VPS).

6. Troubleshooting

Symptom	Action
"Server problem" in the application	`curl https://api.<domain>/health`; `journalctl -u nhill-server`
Session not remembered on Linux	install/unlock GNOME Keyring or KWallet
Session not remembered on Windows or Android	check that "Remember me" is ticked (Settings › My account › This device); Android: set a screen lock; a "secure storage unavailable" message appears if the system refuses to save
"Forgot your password?": no code received	server: `NH_EMAIL_TRANSPORT=smtp`? relay error in `journalctl -u nhill-server`; user: spam folder; an employee of a single establishment receives the instruction to contact their administrator, not a code
"Update required"	install the version requested by the server
Android: "App not installed"	an APK with another signature is present: uninstall it
Windows: application does not start	reinstall; check that the antivirus has not quarantined a file
Server refuses to start, `server.env` readable	`chmod 600 /etc/nhill/server.env`
Server: modified migration refused	never edit a published migration; restore the original